

Learning with ELK

Elasticsearch • Logstash • Kibana • Beats

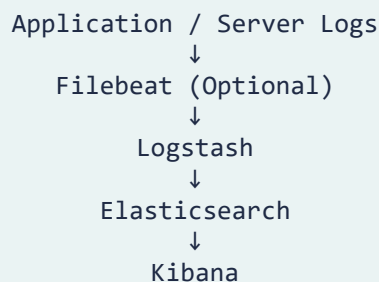
Notes · 31 July 2026

What is ELK Stack?

ELK Stack is a collection of three open-source tools used to collect, process, store, search, analyze, and visualize logs generated by applications, servers, containers, and cloud infrastructure.

ELK = Elasticsearch + Logstash + Kibana

Later, Elastic introduced Beats, so the full stack is often called the Elastic Stack.



When we say Elasticsearch is a distributed search and analytics engine, it means:

- Data is stored on multiple servers (nodes).
- Each server holds part of the data.
- All servers work together as a single Elasticsearch cluster.
- If one server fails, others can continue serving requests (if replicas exist).

1. Elasticsearch

What is Elasticsearch?

Elasticsearch is a distributed search and analytics engine where all processed logs are stored.

Think of it as a high-speed database for logs.

Instead of using SQL like MySQL, Elasticsearch stores data as JSON documents.

```
{
  "timestamp": "2026-07-31T10:00:00",
  "level": "ERROR",
  "service": "payment-api",
  "message": "Database connection failed"
}
```

What Elasticsearch does

- Stores logs
- Searches millions of logs quickly

- Filters logs
- Aggregates data
- Performs analytics
- Indexes documents

Core Concepts

1. What is an Index in Elasticsearch?

An Index is a collection of related documents. It is similar to a database in MySQL or MongoDB.

SQL	Elasticsearch
Database	Index
Table	(No exact equivalent)
Row	Document
Column	Field

Example: logs-2026.07.31 — this index stores all logs generated on 31 July 2026.

2. What is a Document in Elasticsearch?

A Document is a single record stored inside an index. It is stored in JSON format.

```
{
  "user": "Anil",
  "status": "success",
  "ip": "192.168.1.10",
  "timestamp": "2026-07-31T10:20:00"
}
```

This entire JSON object is one document.

3. What is a Field in Elasticsearch?

A Field is an individual key inside a document.

```
{
  "user": "Anil",
  "status": "success",
  "age": 24
}
```

4. What are Shards in Elasticsearch?

A Shard is a smaller piece of an index. Instead of storing all data on one server, Elasticsearch divides the index into multiple shards.

Example — an index with 300 GB of data is split as:

logs-index (300 GB)

Shard 1 → 100 GB

Shard 2 → 100 GB

Shard 3 → 100 GB

Each shard can be stored on a different node.

Why use shards?

- Faster search
- Parallel processing
- Easy scaling
- Large data handling

Note: if you search for “ERROR”, all three shards search simultaneously and return results much faster.

5. What is a Replica in Elasticsearch?

A Replica is a copy of a primary shard.

Primary Shard 1

↓

Replica Shard 1

If the primary server crashes, Elasticsearch automatically uses the replica.

Easy Way to Remember

Concept	Easy Meaning
Index	Database
Document	One JSON record
Field	One key/value inside a document
Shard	One piece of an index
Replica	Copy of a shard
Cluster	Group of Elasticsearch nodes
Node	One Elasticsearch server
Distributed	Data spread across multiple nodes working together
Scalable	Add more nodes as data grows
Near Real-Time	New data becomes searchable in about 1 second

Beats

1. What are Beats?

Beats are lightweight agents installed on servers to collect data and send it to Logstash or directly to Elasticsearch. Instead of Logstash reading log files from every server, Beats run on each machine and forward the data.

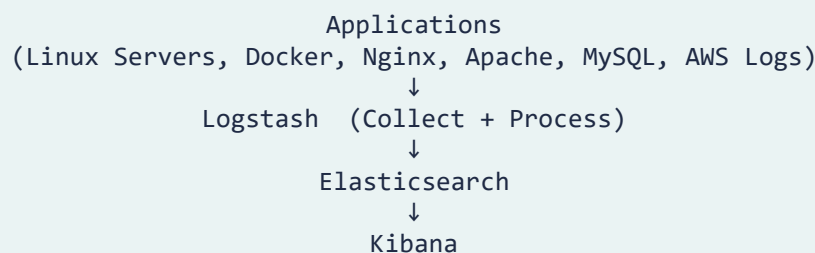
Types of Beats

Beat	Purpose
Filebeat	Collect log files
Metricbeat	Collect metrics
Packetbeat	Collect network traffic
Winlogbeat	Windows Event Logs
Auditbeat	Linux audit logs
Heartbeat	Uptime monitoring

2. Logstash

Logstash is a server-side data processing pipeline that collects data from multiple sources, processes or transforms it, and sends it to Elasticsearch or other destinations.

- Applications generate logs.
- Logstash collects them.
- Logstash formats and cleans the logs.
- Elasticsearch stores them.
- Kibana displays them.



Main Components of Logstash

1. Input

The Input plugin collects data from different sources. Examples:

- Log files
- Docker logs
- Jenkins logs
- Linux syslogs

- Beats (Filebeat, Metricbeat)
- Kafka
- RabbitMQ
- TCP/UDP
- HTTP
- MySQL

Application Log
↓
Input Plugin

2. Filter

The Filter plugin modifies or enriches the data. It can:

- Parse logs
- Remove unnecessary fields
- Rename fields
- Convert data types
- Add new fields
- Extract IP addresses
- Extract usernames

Common filter plugins:

Plugin	Purpose
grok	Extracts structured data from text.
mutate	Renames, removes, or changes fields.
date	Converts timestamps.
geoip	Adds location information from an IP address.
json	Parses JSON logs.

3. Output

The Output plugin sends the processed data to a destination. Common outputs:

- Elasticsearch
- File
- Kafka
- AWS S3
- Email
- Stdout (console)

Sample Logstash Configuration

```
input {
```

```

file {
  path => "/var/log/app.log"
  start_position => "beginning"
}

filter {
  grok {
    match => { "message" => "%{TIMESTAMP_ISO8601:time} %{LOGLEVEL:level} %{GREEDYDATA:msg}" }
  }
}

output {
  elasticsearch {
    hosts => ["localhost:9200"]
    index => "application-logs"
  }
}

```

3. Kibana

Kibana is the visualization and analytics tool of the Elastic Stack (ELK).

It reads data from Elasticsearch and allows you to search, analyze, and visualize data through dashboards, charts, and graphs.

Kibana is an open-source visualization tool that reads data from Elasticsearch and helps users search logs, create dashboards, monitor applications, generate alerts, and analyze data in real time.

Kibana Features

- Discover — Search logs.
- Dashboard — Combine multiple visualizations.
- Visualize — Create pie chart, bar chart, line graph, heat map, data table.

Default Ports

Port	Component / Purpose
9200	Elasticsearch — Client REST API (HTTP). curl http://localhost:9200
9300	Elasticsearch — Internal node-to-node communication
5601	Kibana — http://localhost:5601
5044	Logstash — used by Filebeat
5000	Logstash — TCP or UDP log input
8080	Logstash — HTTP input plugin

End-to-End Pipeline

Application Logs

```
graph TD; A[ ] --> B[Filebeat]; B -- "(port 5044)" --> C[Logstash]; C -- "(port 9200)" --> D[Elasticsearch]; D -- "(port 5601)" --> E[Kibana];
```

↓
Filebeat
↓ (port 5044)
Logstash
↓ (port 9200)
Elasticsearch
↓ (port 5601)
Kibana